

BSC6882: Convexity and variation

Jeffrey West¹, 

¹Integrated Mathematical Oncology Department, H. Lee Moffitt Cancer Center & Research Institute

 @mathoncbro

1 Motivation

Suppose that a biological system is well-characterized by a mathematical model describing the relationship between some inputs, x , and outputs, $f(x)$. In the examples below, we will consider an input dose, x , and a response, $f(x)$, e.g. the number of tumor cells killed.¹ We begin by reviewing a formal definition of Jensen's Inequality, which compares the expected value of a function to the function evaluated at the expected value^{1,2}. For a convex function $f(x)$, the inequality can be written:

$$\mathbb{E}(f(x)) > f(\mathbb{E}(x)). \quad (1)$$

The inequality is determined by the geometric shape of $f(x)$. Conversely, if $f(x)$ is concave the inequality will be flipped:

$$\mathbb{E}(f(x)) < f(\mathbb{E}(x)). \quad (2)$$

It is also possible that the function $f(x)$ is linear, thus exactly equivalent:

$$\mathbb{E}(f(x)) = f(\mathbb{E}(x)). \quad (3)$$

Thus, the payoff function representing a system determines whether input variation can maximize $f(x)$ (convex) or minimize $f(x)$ (concave). We note that $f(x)$ can be defined as a positive (favorable) function of benefit (as it is above) or a negative (harmful) function which will flip the interpretation of convexity/concavity without loss of generality. Consider the two example functions in figure 1.

1.1 Convexity in dose response

To further gain intuition, consider the example of dose-effect in medicine. We compare the effect of a dose, x , to an alternative dosing schema of 120% x followed by a lower dose of 80% x . If dose response is convex, we expect:

$$\frac{f(x + \Delta) + f(x - \Delta)}{2} > f(x) \quad (4)$$

¹These notes were adapted from the forthcoming textbook "Applied Antifragility in Natural Systems, with special thanks to Ari Barnett, Cristian Axenie, Roman Bauer and Oliver López Corona. There is nice link between convexity (or concavity) and antifragility, which is beyond the scope of these notes. Briefly, antifragility is defined as the benefit of input distribution unevenness, volatility, or perturbations attributed to the non-linearity (e.g. curvature) of the system's payoff function. In such cases where perturbing the input increases the resulting output, we use the term antifragility. Ecological antifragility considers the biological system in isolation (without respect to environmental perturbations, evolution by natural selection, or other external disturbances) to quantify the response to perturbations.

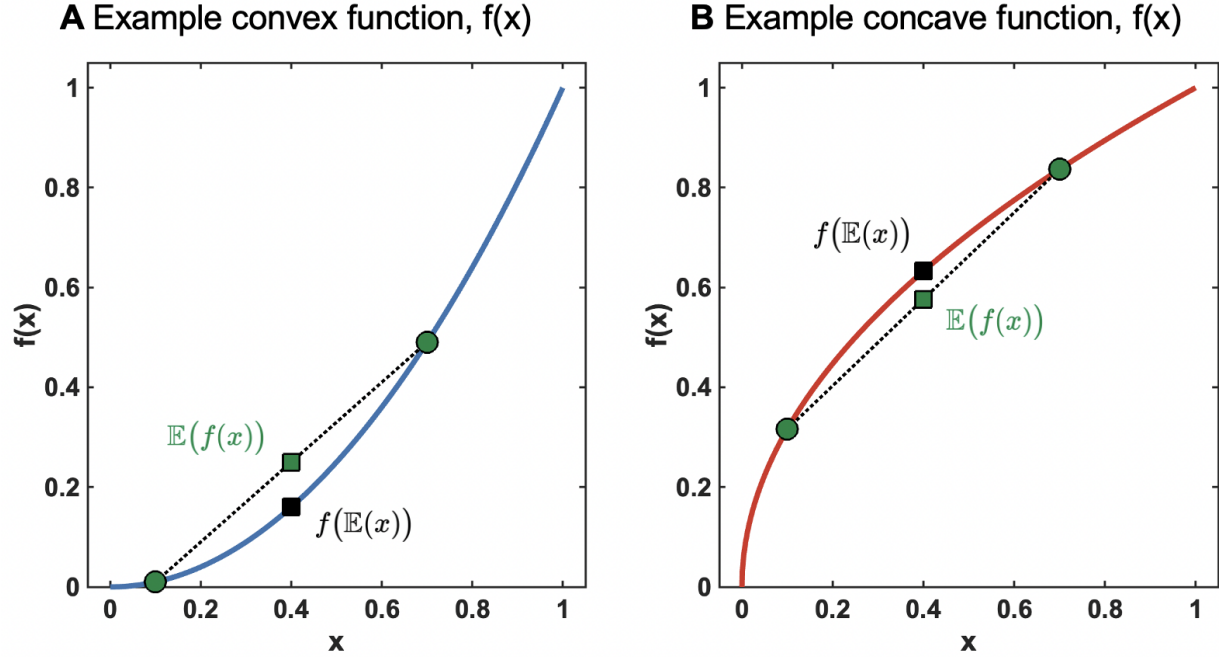


Figure 1. (A) Example convex function, where $f(x) = x^2$. Response to a high and low dose (e.g. eqn. 4) where $x = 0.4$ and $\Delta = 0.3$ is shown by green circles and compared to a constant dose of $x = 0.4$ (black square). The average of the high and low dose, $\mathbb{E}(f(x))$ (green square) outperforms the response to the average dose, $f(\mathbb{E}(x))$. (B) Example concave function, where $f(x) = x^{(1/2)}$, where the opposite conclusion is drawn: $\mathbb{E}(f(x)) < f(\mathbb{E}(x))$.

The left-hand side is the average (indeed, the expectation, $\mathbb{E}(f(x))$) of several dose effects, while the right-hand side is the function evaluated at the average dose, or $f(\mathbb{E}(x))$. This is illustrated in figure 1A, where the right-hand side is marked with a black square while the left-hand side is marked with a green square, denoting the average of the high and low doses (green circles). If dose-response is concave, the inequality is flipped, and the black square is above the green square.

1.2 Discrete case

This simple example can be extended to weigh the effect of the two uneven doses by the time spent on the dose:

$$\lambda f(x + \Delta) + (1 - \lambda)f(x - \Delta) > f(x), \quad (5)$$

where $0 \leq \lambda \leq 1$. Previously, equation 4 is a special case, for $\lambda = 0.5$. Thus, it's straightforward to arrive at the following general definition of antifragility:

$$\sum_i^N \lambda_i f(x_i) > f\left(\sum_i^N \lambda_i x_i\right) \quad (6)$$

where each λ_i value is a weight such that $\lambda_i \in [0, 1]$ and $\sum_i \lambda_i = 1$.

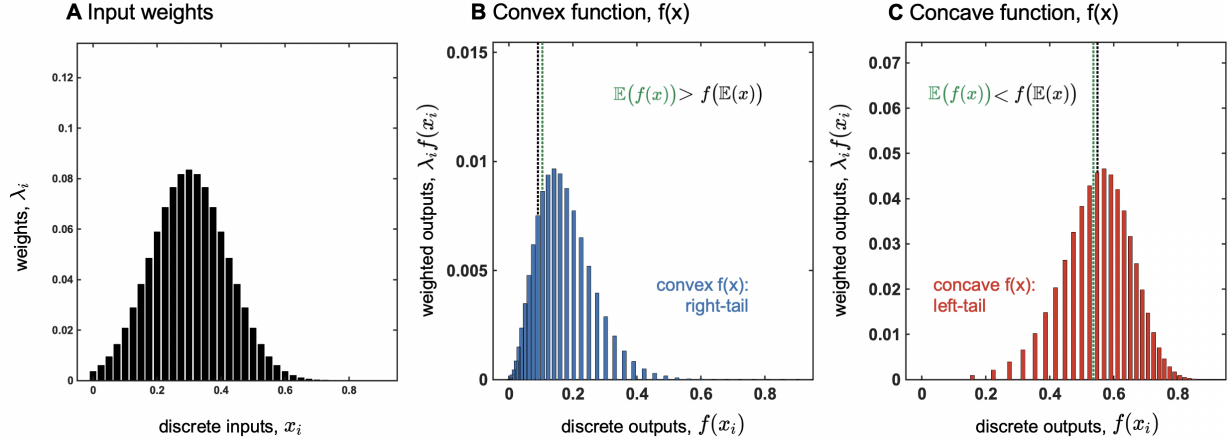


Figure 2. (A) Consider a discrete random variable, $\mathbf{X}$, which has discrete values $\mathbf{X} = (x_1, x_2, \dots, x_N)$ and corresponding weights $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_N)$. Input weights are normally distributed with mean $\mu = 0.3$ and standard deviation $\sigma = 0.12$. (B, C) We plot the distribution of weighted outcomes, $\lambda_i f(x_i)$ for the corresponding convex (B) and concave (C) functions in figure 1. For convex functions (B), the right-tail drives the mean of the outcome distribution up so that $\mathbb{E}(f(x)) > f(\mathbb{E}(x))$. For concave functions, the left-tail drives the mean down, so that $\mathbb{E}(f(x)) < f(\mathbb{E}(x))$.

An example is shown in figure 2A, where we have a set of discrete input values, $\mathbf{X} = (x_1, x_2, \dots, x_N)$ with corresponding weights $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_N)$. Next, we show the distribution of values of $\lambda_i x(x_i)$ in figure 2B, C for the convex and concave functions in the previous figure, respectively. We make several important observations. Firstly, again the sign of the Jensen's inequality is determined by the curvature (convex or concave) of the function $f(x)$. Secondly, while the input distribution of weights, λ , is normally distributed, the output distributions are skewed with a tail to the right (convex) or left (concave). These tails drive the $\mathbb{E}(f(x))$ toward the tail (shown in green dashed line) when compared with the $f(\mathbb{E}(x))$ (black dashed line).

This analysis motivates moving from a discrete distribution of weights to a continuous distribution, which we will discuss in the next chapter. Discrete λ_i will be replaced by continuous probability distribution functions and convolutional methods are employed.

2 Probability convolution for convex and concave functions

The previous chapter has illustrated the connection between the convexity of the payoff function, $f(x)$, and the outcome benefit to input variation. Here, we extend this idea to illustrate the connection between convexity and statistical properties of the distribution of $f(x)$. There are many biological systems for which it is difficult to measure the system's response function. Alternatively, the system's payoff function may indeed be well-characterized but external signals or noise introduce additional, unpredictable nonlinearities.

We begin with a simple example function and use probability convolution methods to derive the exact expression of this outcome distribution of $f(x)$, given some input probability distribution of x . These probability distribution functions (p.d.f.) will have associated properties depending if the

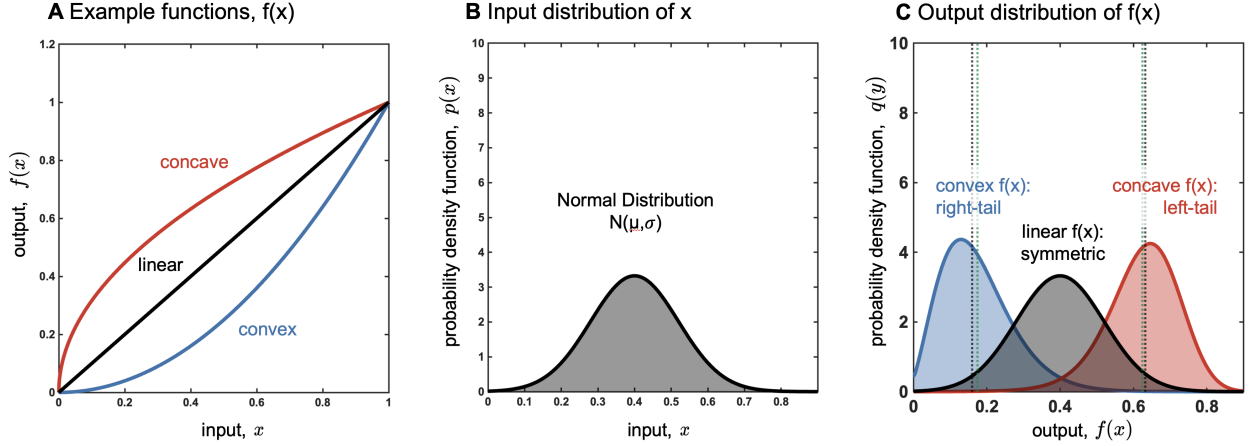


Figure 3. (A) Three example functions are shown. Blue: convex function where $f(x) = x^2$. Red: concave function $f(x) = x^{\frac{1}{2}}$. Black: linear function where $f(x) = x$. (B) Consider a continuous random variable, $\mathbf{X}$, which has an associated probability density function that is normally distributed with mean $\mu = 0.4$ and standard deviation $\sigma = 0.12$. (B, C) We plot the probability density function representing the distribution of outcomes, $f(x)$ for the corresponding convex (blue), concave (red), and linear (black) in A. For convex functions (B), the right-tail drives the mean of the outcome distribution up so that $\mathbb{E}(f(x)) > f(\mathbb{E}(x))$. For concave functions, the left-tail drives the mean down, so that $\mathbb{E}(f(x)) < f(\mathbb{E}(x))$. For linear functions, the outcome distribution is symmetric (no tail).

underlying function is convex, concave, or linear. Consider the following function:

$$f(x) = x^c \quad (7)$$

where c is a constant. As seen in figure 3A, the shape of this function can be convex ($c > 1$; blue), concave ($c < 1$; red), or linear ($c = 1$; black). Let $p(x)$ be the probability density function describing the input distribution of x . For example, figure 3B shows a normal distribution with corresponding μ, σ : $p(x) = N(\mu, \sigma)$:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}. \quad (8)$$

The probability density function of the outcome, $f(x)$ can be determined by convolution. Let X be a random variable (input) and let $y(X) = f(X)$, then the outcome distribution of $f(x)$ is given by:

$$P(y(a) \leq Y < y(b)) = \int_a^b p(x) dx \quad (9)$$

$$= \int_{y(a)}^{y(b)} p(x(y)) \left| \frac{dx}{dy} \right| dy \quad (10)$$

where the argument inside the integral is the p.d.f. of $f(x)$, which we denote as $q(y)$:

$$q(y) = p(x(y)) \left| \frac{dx}{dy} \right|. \quad (11)$$

For the example function $f(x)$ given above, both terms of the integrand can be found analytically:

$$x(y) = y^{\frac{1}{c}}, \quad (12)$$

and

$$\left| \frac{dx}{dy} \right| = \frac{1}{c} y^{\frac{1-c}{c}}. \quad (13)$$

which gives:

$$q(y) = \frac{y^{\frac{1-c}{c}}}{\sigma c \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{y^{\frac{1}{c}} - \mu}{\sigma} \right)^2}. \quad (14)$$

The key parameter is c , which determines the tailedness of the outcome distribution, as shown in figure 3C. Convex functions ($c > 1$) are associated with right-tails which cause an increase in the mean of the distribution, such that $\mathbb{E}(f(x)) > f(\mathbb{E}(x))$. Concave functions ($c < 1$) are associated with a left-tail leading to the opposite conclusion: $\mathbb{E}(f(x)) < f(\mathbb{E}(x))$. The implications for systems where the system response function $f(x)$ is unknown are important. Because of the nature of probability distributions, everything left-tailed must be concave while everything right-tailed must be convex³. Thus, it's possible to determine the benefit (or harm) from input variation through knowledge of the tailedness of the outcome distribution, without knowledge of the underlying functional response, $f(x)$. In the following sections, we illustrate an example of this approach in medicine.

In figure 3C, we can directly compare $\mathbb{E}(f(x))$ (green line) and $f(\mathbb{E}(x))$ (black line) as follows:

$$f(\mathbb{E}(x)) - \mathbb{E}(f(x)) = \mu^c - \int_{-\infty}^{\infty} y q(y) dy \quad (15)$$

The differences are shown in figure 3C, confirming that this difference is negative for convex functions, positive for concave functions and zero for linear functions.

2.1 Caveats and considerations

It's important to note that each of these formulations implicitly assumes that the effect of each input (e.g. each dose administered) is **independent and additive**. Thus far, we deal with cases where the payoff function, $f(x)$, is well known and any external disturbances or environmental signals which alter $f(x)$ are negligible. Without loss of generality, $f(x)$ can describe either benefit or harm to a system, and thus we can flip the inequality in our definition of antifragility to maximize benefit or minimize harm, respectively.

3 Convexity in cancer treatment is multiplicative

Next, we illustrate a simple example of convexity applied to medicine. Consider a population of tumor cells under treatment with initial size n_0 and growth rate $\gamma(x)$ the final size, n , is given by:

$$n(T) = n_0 \exp(\gamma(x)T) \quad (16)$$

where $\gamma(x)$ is the rate of growth of the tumor under treatment with dose x over a time interval of T . The quantity within the parenthesis is known as the log-kill effect, which we denote by the parameter $\beta(x) = \gamma(x)T$. Thus, the effect of multiple doses can be predicted via a summation of the log-kill effect.

$$n = n_0 \exp(\beta(x) + \beta(x)) \quad (17)$$

Next, we assess the effect of variation in the input dose, x , by comparing an even (non-volatile) dosing scheme to an uneven (volatile) dosing scheme. Consider a cycle of treatment:

$$\text{Treatment Cycle} = \{\bar{x} + \Delta, \bar{x} - \Delta\} \quad (18)$$

where $\bar{x}$ represents the dosing schedule's mean dose value and Δ represents the dosing schedule's variance. In this way, all treatment schedules have identical dose mean, enabling us to compare second-order effects that are due to alterations in variance only. We term a dosing schedule "even" if there is no variance ($\Delta = 0$) or "uneven" if the variance is non-zero ($\Delta > 0$).

$$\begin{aligned} n_{\text{even}} &= n_0 \exp(\beta(x) + \beta(x)) \\ n_{\text{uneven}} &= n_0 \exp(\beta(x + \Delta) + \beta(x - \Delta)) \end{aligned}$$

As mentioned above, the definition of antifragility (e.g. eqn. 4) requires that the effect of each input is additive and independent. The dose-dependent log-kill rate, $\beta(x)$, satisfies both conditions. Thus, it is useful to define a metric of fragility, F , to compare the log-kill rate of a tumor under even ($\sigma = 0$) dosing or under uneven ($\sigma > 0$) dosing:

$$F(x, \Delta) = \beta(x + \Delta) + \beta(x - \Delta) - 2\beta(x). \quad (19)$$

If uneven treatment protocols result in maximizing tumor regression then it will result in $F < 0$ and we term this situation antifragile. Conversely, if even dosing protocols result in maximizing tumor regression, $F > 0$, it is fragile.

Rearranging some terms, it's straightforward to show the following relation:

$$F = \ln \left(\frac{n_{\text{uneven}}}{n_{\text{even}}} \right) \quad (20)$$

Or, more conveniently:

$$n_{\text{uneven}} = n_{\text{even}} \exp(F) \quad (21)$$

Therefore, fragility, F , is interpreted as the log-kill gain (or loss) of switching to an uneven treatment schedule, compared to baseline even dosing. Patients will benefit (i.e. maximizing tumor kill) by switching to an uneven dosing protocol when $F < 0$. The appropriate measure of convexity is the exponential growth rate of tumors as a function of dose delivered, $\beta(x)$, and therefore precise characterization of the convexity (or concavity) of this function is warranted. Examples are shown in figure 4.

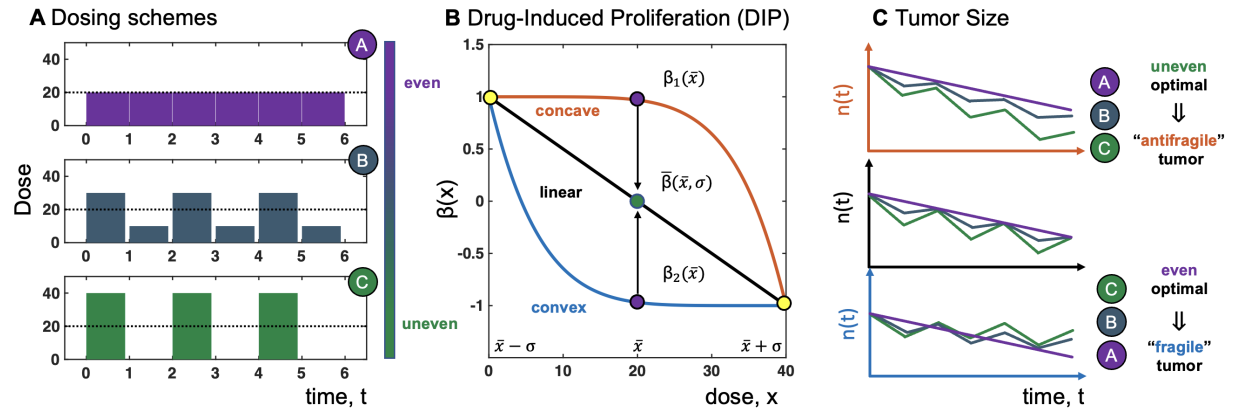


Figure 4. Dose response convexity predicts optimal treatment schedule. A) Schematics of even (purple) and uneven (green) treatment schedules with the same cumulative dose delivered. B) Examples of dose-response curves with differing convexity. The green dot indicates the difference between the average $\beta(x)$ for two doses versus the $\beta(x)$ at the average dose delivered. C) Predicted tumor size based on each dose-response curve. When the dose-response is concave, more tumor is eliminated by uneven treatment cycles, so the tumor is antifragile. Conversely, if the dose-response is convex, then even treatment cycles are preferable and the tumor is fragile. Figure adapted from reference⁴.

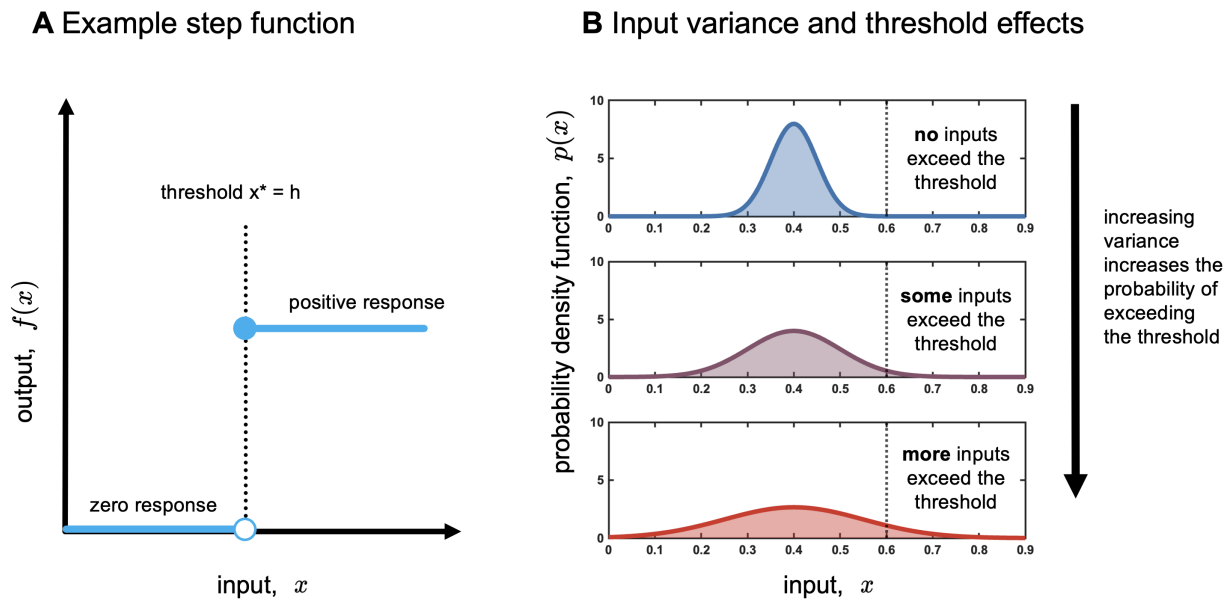


Figure 5. Relationship between input distribution variance and threshold effects. Assume an input distribution (e.g. Normal distribution with associated mean μ and variance σ and some threshold value, $x^* = h$ such that $h > \mu$. As variance increases, the likelihood of observing events that exceed the threshold also increases.

4 Threshold effects lead to antifragility

Finally, we illustrate how systems with a threshold effect may benefit from an increase in system variability. A step function (figure 5A) has no response below a certain threshold value, h . A step function experience an increase in mean output with an increase in variance (it acts is convex). Consider the effect of increasing the input distribution variance in figure 5B, where low variance leads to zero probability (figure 5B, top), but high variance extends the right tail to exceed the threshold value.

This scenario is widely applicable in natural systems. For example, BT crops that have been genetically modified to produce a constant level of pesticide still perform worse when compared to occasional, manual interventions by farmers⁵. This is because the pesticide self-production by these crops is lower than the threshold required to affect pests. Occasion intervention dramatically increases the variance (e.g. weekly, or monthly) but each intervention exceeds the required threshold for response. Other examples in nature include stochastic resonance, where stochastic noise increases the signal's response to rising above the threshold of detection⁶. Observing the concavity of the dose-response curve in ovarian cancer has shown promise in designing treatment scheduling with reduced variance, while at the same time lowering the cumulative dose⁷.

References

1. Jensen, J. L. W. V. *et al.* Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta mathematica* **30**, 175–193 (1906).
2. Kerbel, R. S. & Kamen, B. A. The anti-angiogenic basis of metronomic chemotherapy. *Nature Reviews Cancer* **4**, 423–436 (2004).
3. Axenie, C. *et al.* Antifragility in complex dynamical systems. *npj Complexity* **1**, 12 (2024).
4. Pierik, L., McDonald, P., Anderson, A. R. & West, J. Second-order effects of chemotherapy pharmacodynamics and pharmacokinetics on tumor regression and cachexia. *Bulletin Mathematical Biology* **86**, 47 (2024).
5. Taleb, N. N. & West, J. Working with convex responses: Antifragility from finance to oncology. *Entropy* **25**, 343, DOI: [10.3390/e25020343](https://doi.org/10.3390/e25020343) (2023).
6. Gammaitoni, L., Hänggi, P., Jung, P. & Marchesoni, F. Stochastic resonance. *Reviews Modern Physics* **70**, 223 (1998).
7. Strobl, M. A. *et al.* To modulate or to skip: De-escalating parp inhibitor maintenance therapy in ovarian cancer using adaptive therapy. *Cell Systems* **15**, 510–525 (2024).